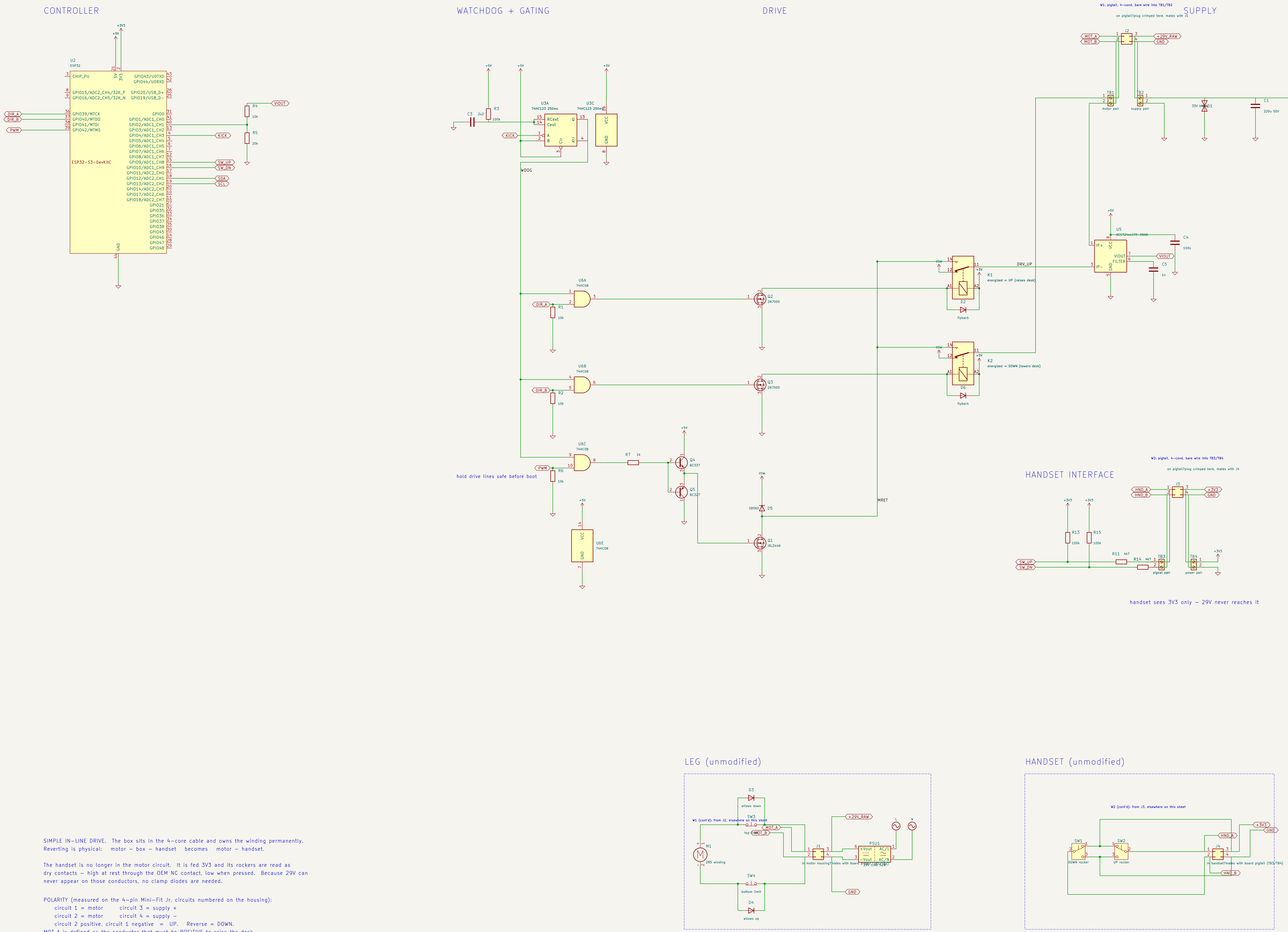

SIMPLE IN-LINE DRIVE. The box sits in the 4-core cable and owns the winding permanently.
Reverting is physical: motor – box – handset   becomes   motor – handset.

The handset is no longer in the motor circuit. It is fed 3V3 and its rockers are read as dry contacts – high at rest through the OEM NC contact, low when pressed. Because 29V can never appear on those conductors, no clamp diodes are needed.

POLARITY (measured on the 4-pin Mini-Fit Jr, circuits numbered on the housing):
    circuit 1 = motor   circuit 3 = supply +
    circuit 2 = motor   circuit 4 = supply –
    circuit 2 positive, circuit 1 negative = UP.   Reverse = DOWN.
MOT_A is defined as the conductor that must be POSITIVE to raise the desk (circuit 2). MOT_B is circuit 1.

K1 / K2 reproduce the OEM rest behaviour:
    both off -> both winding ends on VSW = BRAKE
    K1 on -> MOT_B to return, MOT_A stays positive = UP
    K2 on -> MOT_A to return, MOT_B stays positive = DOWN
No combination of relay states can short the supply; there is no shoot–through failure mode.

SEQUENCING: relays change state only with Q1 off and current decayed (~50ms). On a watchdog trip Q1 turns off in microseconds while the relays take milliseconds to release, so the contacts are cold–switched even in a fault. Loss of drive is coast, which is safe because the spindle is self–locking and holds the load unpowered.

U5 sits in the motor leg so it reads current during freewheel as well as during the PWM on–time, which a low–side shunt would not. ISENSE must be on ADC1 (ADC2 dies with WiFi).
R1 / R2 / R6 must sit on non–strapping GPIOs.

Box owns the winding; unplug it to revert

Sheet: /
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Title: Slangerup desk retrofit – in–line drive

Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 1/1